Earth and Planetary Science Letters, 97 (1990) 54–64
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[DT]

Time resolution of geologic events on the Keweenaw Peninsula and implications for development of the Midcontinent Rift system

D.W. Davis [1] and J.B. Paces [2,*]

[1] *Jack Satterly Geochronology Laboratory, Royal Ontario Museum, 100 Queen's Park, Toronto, Ont. M5S 2C6 (Canada)*
[2] *Department of Geological Engineering, Geology and Geophysics, Michigan Technological University, Houghton, MI 49931 (U.S.A.)*

Received June 21, 1989; revised version received October 18, 1989

Three flood-type lava flows from the Keweenaw Peninsula in upper Michigan have been precisely dated, using U-Pb analyses of zircon and baddeleyite. Zircons were extracted from pegmatoid segregations within two basaltic flows. Ages from the Copper City Flow, near the base of the Portage Lake Volcanics, and the Greenstone Flow, near the top, are 1096.2 ± 1.8 Ma and 1094.0 ± 1.5 Ma, respectively (95% confidence errors). U-Pb analyses of baddeleyite from these samples are in general agreement with the zircon results although some fractions appear slightly older, possibly due to ^{222}Rn loss. The age of an andesite flow from the Lake Shore Traps, 1000 m above the Portage Lake Volcanics and within the Copper Harbour Conglomerate, is 1087.2 ± 1.6 Ma from analyses of zircon. This gives an age for sedimentation within the Copper Harbor Conglomerate. These ages, along with stratigraphic data, indicate a subsidence rate of $1.3^{+1.8}_{-0.6}$ mm/y during eruption of the Portage Lake Volcanics and 0.2 ± 0.1 mm/y (1σ errors) for the transitional period from flood basalt eruption to clastic sedimentation. Magma eruption rates were at least 0.02–0.05 km^3/y during emplacement of the Portage Lake Volcanics. Rates as high as 0.2–0.6 km^3/y are inferred from regional stratigraphic data and currently available precise dates from throughout the Lake Superior basin. These values are comparable to the high rates of magma production estimated for Phanerozoic continental flood basalt provinces. Analyses of three detrital zircon grains from a sandstone at the base of the Copper Harbor Conglomerate give ages from 1101 to 1106 Ma. A fraction consisting of 29 detrital zircon grains has an age of 1104 ± 2 Ma indicating that sediments within the Copper Harbor Conglomerate were probably derived almost entirely from lower Keweenawan rocks.

Since the Portage Lake Volcanics are paleomagnetically normal, the age of the Keweenawan paleomagnetic polarity reversal is now bracketed between 1096.2 ± 1.8 Ma and 1097.6 ± 3.7 Ma, the age of a reversed flow within the Osler Group. From this and other studies it appears that apparent polar wander rates are different for reversed and normal poles and that normal and reversed paleomagnetic poles of comparable age have different geographic positions. This suggests that the asymmetry between reversed and normal Keweenawan poles for igneous rocks may be due to a non-geocentric dipole.

1. Introduction and geology

The Keweenaw Peninsula in upper Michigan has been considered the type area for rocks which form part of a major, failed Proterozoic rift known as the Midcontinent Rift system [1,2], in the region of Lake Superior. This failed rift structure can be traced geophysically for at least 2000 km.

Historical attempts to date rocks associated with the Midcontinent Rift system are summarized by Van Schmus et al. [3]. Efforts to date some of the intrusions, using K-Ar and Ar-Ar, gave ages which are too old (probably due to excess argon), while most Rb-Sr ages on Keweenawan rocks appear to be too young (possibly due to the effects of regional, low-temperature hydrothermal alteration). The application of U-Pb zircon geochronology has led to a considerable reduction in the perceived time span of igneous activity. Results presented by Silver and Green [4] indicated that the bulk of igneous activity was 1110 ± 10 Ma in age. Although these data were never published in detail, the effects of secondary lead loss presumably prevented the resolution of individual ages within the time span of rift development. The introduction of the zircon abrasion technique [5] combined with high sensitivity, ex-

* Present address: Department of Geological Sciences, University of Tennessee, Knoxville, TN 37996-1410, U.S.A.

0012-821X/90/$03.50 © 1990 Elsevier Science Publishers B.V.

tended geometry mass spectrometers and analytical blanks on the order of a few picograms have enabled age determinations precise to 1–2 Ma on zircon fractions consisting of only a few grains. The thick lava piles presently exposed in the Lake Superior region represent relatively rapid accumulation of conformable flood basalt lava flows within subsiding rift basins. Resolution of individual ages within such stratigraphic successions using U-Pb analyses of zircon and/or baddeleyite provides an excellent test of the accuracy of the age determination technique. If successful, high-precision dating would not only provide a basis for absolute age correlation of events in different parts of the Midcontinent Rift system, but also would provide age information allowing estimation of the rates of physical volcanological and sedimentological processes.

In addition, the paleomagnetism of rocks associated with the Midcontinent Rift system has been extensively studied and has provided the most temporally continuous and internally consistent part of the North American Precambrian polar wander path. Precise time calibration of this path may help to determine whether it is due to continental motion and, if so, could provide information on rates of continental drift during the Precambrian.

Recently, several precise zircon age determinations have been carried out on units in different parts of the rift system which help to establish the timing of the polar wander curve. The oldest is for

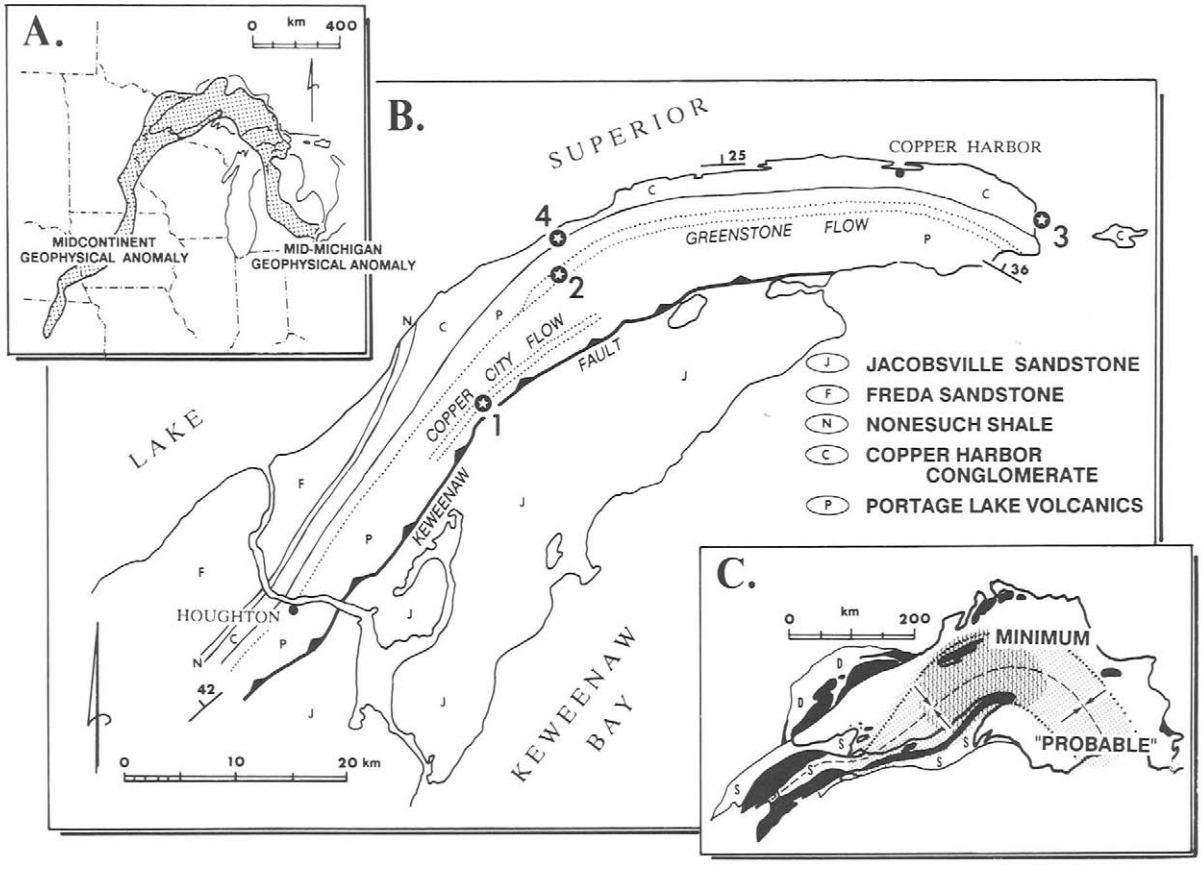


Fig. 1. A. General location map showing the geophysical expression (gravity and magnetic high) of the Midcontinent Rift (shaded). B. Generalized geological map of the Keweenawan Peninsula showing sample locations (sample numbers are keyed to Table 1). C. Extent of Midcontinent Rift rocks exposed in the Lake Superior Region. Black areas are predominantly volcanics, D = Duluth Complex intrusive rocks; S = Oronto Group and Bayfield Group rift-fill sediments. Central rift graben is defined by the Keweenawan and Isle Royale faults (dotted lines) on either side of the axis of the Lake Superior Syncline (dashed line). "Minimum" and "Probable" shaded areas within the graben refer to areas used in volume calculations.

the Logan sills which have a zircon age of 1109^{+4}_{-2} Ma [6]. The youngest is for a felsic porphyry from Michipicoten Island which has a zircon age of $1086.5^{+1.3}_{-3.0}$ Ma [7]. These units lie approximately at the end points of the paleomagnetic polar wander curve for Keweenawan igneous rocks [8]. Midcontinent Rift activity spanned at least one polarity reversal, with stratigraphically lower units exhibiting reversed polarity and upper units exhibiting normal polarity. Evidence for multiple reversals is found at Mamainse Point [9,10]. However, the R–N paleomagnetic reversal is well documented in several sequences and has been used for correlation of units in different parts of the rift system. The youngest reversed unit which has been precisely dated is from the Osler Group and has a zircon age of 1097.6 ± 3.7 Ma [6]. Up to the present work, precise dating of these three paleomagnetic poles had been consistent with their relative pole positions and indicated an apparent polar wander rate of about 14 cm/y [7].

The present study utilizes samples from the Portage Lake Volcanics and Copper Harbor Conglomerate in the Keweenaw Peninsula of Upper Michigan (Figs. 1 and 2). Age determinations on these samples are important because they span the stratigraphic interval between the older, reversed polarity Osler Group rhyolite and the younger, normal polarity Michipicoten Island samples (i.e., regional stratigraphy of Green [11]). The Portage Lake Volcanics, exposed on both the southeast and northwest limbs of the Lake Superior syncline (Keweenaw Peninsula and Isle Royale respectively) consist of a 3–5 km thick pile (fault-truncated base) of predominantly olivine tholeiite flood-type lavas with only minor thicknesses of interflow sedimentary and felsic volcanic units. Feeder-dyke complexes for the Portage Lake Volcanics have not been recognized and are presumably located within the central rift graben between the Keweenaw and Isle Royale Faults (Fig. 1). The top of the Portage Lake Volcanics is conformably overlain by, and in places is intercalated with, terrigenous clastic sediments of the Copper Harbor Conglomerate. The 2 km thick sequence of conglomerates, sandstones, siltstones and interbedded lava flows has been interpreted as a prograding alluvial fan complex [12] deposited within the rift after the main stage of volcanism ceased. Paleocurrent directions indicate movement of the sediment from the margin to the center of the basin. Basin subsidence occurred continuously during the transition period between volcanic accumulation and clastic sedimentation [13]. This sequence of rocks therefore records a major transition in Midcontinent Rift evolution from dominantly volcanic to sedimentary rift-filling material. In addition, all rocks within the Keweenaw Peninsula section exhibit normal magnetic polarity. Paleomagnetic and stratigraphic relationships indicate that this section represents the youngest flood basalt-filled portion of the Midcontinent Rift system exposed in the western Lake Superior region [14].

Neither the basaltic flow rocks which make up the bulk of the Portage Lake Volcanics nor the aphyric rhyolites found near its base are normally considered likely to contain zircon. However, some of the basalt flows are exceptionally thick (100–400 m) and contain lenses of differentiated micro-

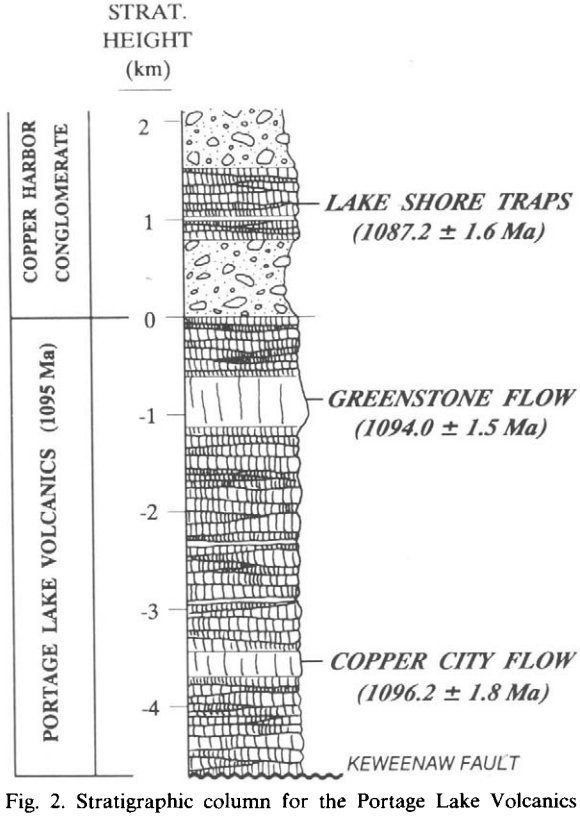


Fig. 2. Stratigraphic column for the Portage Lake Volcanics and a portion of the Copper Harbor Conglomerate exposed on the Keweenawan Peninsula. Positions of geochronology samples are also shown.

pegmatitic material [15]. Pegmatoid lenses consist predominantly of 1–10 mm laths of Na-rich plagioclase and prisms of Fe-rich augite along with interstitial ilmenite, magnetite and apatite, and are often vesicular. These layers represent the crystallized products of the final, volatile-rich liquids remaining after extensive in situ differentiation and are the materials most likely to contain accessory zircon and/or baddeleyite. Pegmatoid rocks were sampled from the Copper City Flow in the lower part of the section and the Greenstone Flow in the upper part (Fig. 2). These two units are separated by about 2800 m.

In order to date the age of deposition of the Copper Harbor Conglomerate, a sample was collected from a feldspar–phyric andesite flow, containing sparse quartz phenocrysts, within a series of conformably interbedded Fe-rich basalts through andesites known collectively as the Lake Shore Traps. The sampled Lake Shore Trap flow is separated from the Greenstone Flow by 700 m of Portage Lake lavas and by 800–1000 m of coarse clastic sediment. A sample of coarse sandstone was also collected from the base of the Copper Harbor Conglomerate directly overlying the Portage Lake Volcanics in order to analyze detrital zircon grains. These should give the age spectrum of the source area and thus help to determine the provenance of the sediments.

2. Methods and results

Approximately 35 kg of each pegmatoid within the Portage Lake Volcanics yielded only a few milligrams of zircon and baddeleyite. Zircon is generally found as clear, fractured prismatic crystals while baddeleyite forms brown, very thin tabular crystals. Extraction and analysis of zircon and baddeleyite followed standard procedures at the Royal Ontario Museum laboratory [16,17]. Hand-picked zircon fractions were strongly abraded to remove surface-correlated secondary lead loss [5]. Baddeleyite crystals could not be abraded due to their thin, wafer-like shape.

Because of the very small zircon yield and the necessity to analyze only crack-free grains, very small fractions, in some cases consisting of only a single grain, were analyzed. This was somewhat offset by the high uranium content of these grains. The andesite from the Lake Shore Traps is somewhat richer in zircon but the grains are generally much lower in uranium than those from the pegmatoid rocks.

A persistant problem with Keweenawan felsic rocks has been the presence of inherited components within the zircon populations. This results from the fact that many of these rocks are the result of partial melting or contamination involving mid-Proterozoic or Archean basement [6,7]. It is therefore necessary to obtain several replicate, concordant analyses to be sure that the age is not biased by inheritance.

Analyses of zircon fractions from each of the three rocks are concordant, or nearly so, and have $^{207}Pb/^{206}Pb$ ages within error or each other. Rock ages were therefore calculated by averaging the $^{207}Pb/^{206}Pb$ ages by fitting a line between the data cluster and the origin of concordia using the program of Davis [18]. Errors in the resulting age are determined almost entirely by the size of the error ellipses assigned to the data points. These are largely a result of uncertainties in fractionation while measuring small samples. The samples are normally run to exhaustion and errors are determined from internal scatter of the isotopic ratios over the course of the run. There is, however, no precise, reliable formula for error assignment and it is to some degree subjective. An

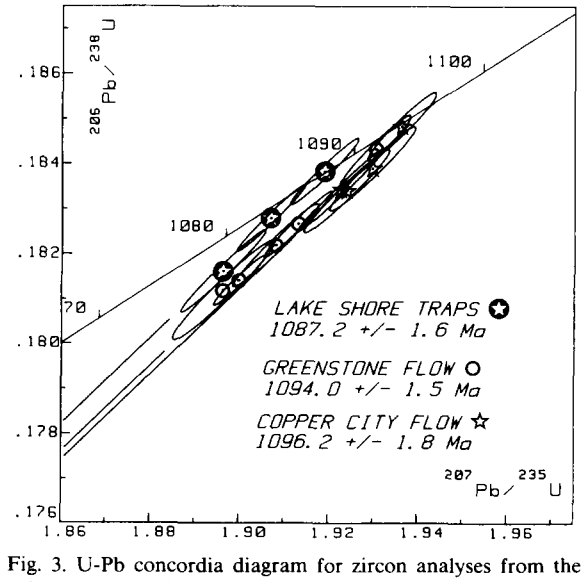


Fig. 3. U-Pb concordia diagram for zircon analyses from the volcanic rocks. The lines are best fit lines to the data clusters for each rock and the origin of the concordia diagram. Error ellipses are at the 1σ level.

TABLE 1

Isotopic data for zircon and baddeleyite from units in the Keweenawan Peninsula

Zircon fractions	Weight (mg)	U (ppm)	Common Pb [a] (pg)	^{207}Pb / ^{206}Pb [b]	^{207}Pb / ^{204}Pb [c]	^{208}Pb / ^{206}Pb [d]	^{206}Pb / ^{238}U [d,e]	^{207}Pb / ^{235}U [d,f]	^{207}Pb / ^{206}Pb Age (Ma) [d,g]
1. Copper City Flow (47°16′34″N, 88°22′51″W)									
1. Zircon Ab	0.004	605	6	0.07611	266	0.529	0.18389	1.9297	1098.0
1 grain					(*)		(1.00)	(0.23)	(1.0)
2. Zircon Ab	0.006	2000	22	0.07744	483	0.709	0.18342	1.9226	1095.5
6 grains					(76)		(1.00)	(0.15)	(1.0)
3. Zircon Ab	0.002	1013	3	0.07598	422	0.366	0.18483	1.9363	1094.5
1 grain					(*)		(0.80)	(0.20)	(0.2)
4. Zircon Ab	0.005	734	5	0.07609	686	0.533	0.18339	1.9239	1097.2
3 grains					(*)		(1.00)	(0.22)	(1.2)
5. Baddeleyite	0.009	466	13	0.07771	275	0.012	0.18291	1.9239	1102.5
15 grains					(777)		(1.00)	(0.23)	(1.9)
6. Baddeleyite	0.023	378	12	0.07657	558	0.012	0.18383	1.9289	1097.6
					(2331)		(0.80)	(0.15)	(1.0)
2. Greenstone Flow (47°23′25″N, 88°17′40″W)									
7. Zircon Ab	0.010	345	4	0.07596	547	0.826	0.18433	1.9306	1094.0
4 clear grains					(*)		(0.80)	(0.20)	(0.3)
8. Zircon Ab	0.001	3637	5	0.07596	577	0.318	0.18266	1.9131	1094.0
1 brown grain					(*)		(0.80)	(0.16)	(1.2)
9. Zircon Ab	0.002	571	5	0.07596	198	0.737	0.18140	1.8999	1094.0
2 clear grains					(*)		(0.60)	(0.14)	(1.9)
10. Zircon Ab	0.004	784	3	0.07596	828	0.930	0.18219	1.9081	1094.0
1 grain					(*)		(0.50)	(0.11)	(1.5)
11. Zircon Ab	0.002	716	3	0.07591	384	0.992	0.18118	1.8963	1092.5
2 cracked grains					(*)		(0.60)	(0.15)	(1.9)
12. Baddeleyite	0.040	123	7	0.07604	372	0.025	0.18341	1.9230	1096.0
					(*)		(0.50)	(0.11)	(1.0)
3. Lake Shore Traps (47°26′00″N, 87°42′52″W)									
13. Zircon Ab	0.086	75	6	0.07571	1055	0.309	0.18383	1.9190	1087.3
21 fragments					(*)		(0.80)	(0.17)	(−0.1)
14. Zircon Ab	0.080	94	5	0.07567	1472	0.267	0.18279	1.9070	1086.1
10 rounded grains					(*)		(0.80)	(0.14)	(0.4)
15. Zircon	0.020	474	8	0.07573	992	0.217	0.18161	1.8963	1088.0
cracked					(*)		(0.80)	(0.14)	(1.2)
4. Copper Harbour Conglomerate (47°24′45″N, 88°17′46″W)									
16. Zircon Ab	0.011	250	8	0.07639	316	0.199	0.18608	1.9599	1105.1
1 grain					(*)		(1.00)	(0.21)	(0.5)
17. Zircon Ab	0.015	244	15	0.07857	230	0.177	0.18336	1.9317	1105.6
1 grain					(517)		(1.00)	(0.21)	(2.0)
18. Zircon Ab	0.020	132	10	0.07678	246	0.156	0.18668	1.9624	1101.2
1 grain					(2023)		(1.00)	(0.18)	(−0.2)
19. Zircon Ab	0.126	113	5	0.07636	2398	0.182	0.18654	1.9639	1104.3
29 grains					(*)		(0.70)	(0.10)	(0.2)

Ab = abraded fraction.

[a] Corrected for 0.117 mole fraction of common lead in the ^{205}Pb spike (except fractions 4, 10, 11, 13, 14 and 19 which were analyzed using a new spike, free of common lead. Includes blank + zircon common lead.

[b] Corrected for fractionation, blank and spike common lead. U blank = 3.8 pg (large column), 0.5 pg (small column); Pb blank < 8 pg (both columns). Pb and U fractionation correction = 0.13%/amu.

[c] Measured value. Numbers in brackets are values corrected for fractionation, blank and common lead in the spike. (*) means all common lead can be considered blank.

[d] Corrected for fractionation, blank and common Pb. Common Pb determined from the Stacey and Kramers model [39].

[e] Numbers in brackets refer to % 2σ error in Pb/U.

[f] Numbers in brackets refer to % 2σ error in $^{207}Pb/^{206}Pb$ age. 0.1% corresponds to an error of 2.0 Ma.

[g] Numbers in parenthesis refer to % discordance for the given $^{207}Pb/^{206}Pb$ age. ^{238}U decay constant = 0.15513×10^{-9} y^{-1}; ^{235}U decay constant = 0.98485×10^{-9} y^{-1}; $^{238}U/^{235}U = 137.88$

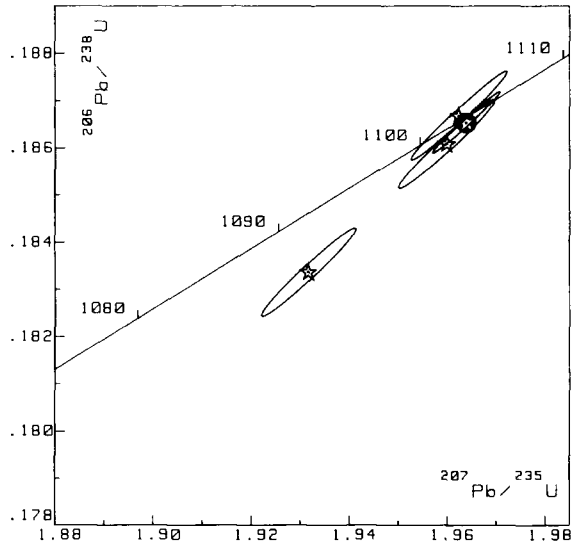


Fig. 4. U-Pb concordia diagram for analyses from single detrital zircon grains in the sandstone from the Copper Harbor Conglomerate. Error ellipses are at the 1σ level.

internal check on the reliability of this procedure is the probability of fit of the data points. If errors are correctly assigned, and one is dealing with a true random distribution, the average probability of fit should be 50%. Values which are excessively above or below this may indicate that assigned errors are too high or too low respectively. If anything, an effort is made to ensure that errors are exaggerated, rather than being underestimated. Errors are normally quoted at the 95% confidence level. Error ellipses are plotted at the 1 σ level in Figs. 3 and 4 to permit visual resolution of the data points.

The age of the Copper City Flow from four zircon fractions is 1096.2 ± 1.8 Ma (66% probability of fit). Five zircon fractions from the Greenstone Flow define an age of 1094.0 ± 1.5 Ma (100% probability of fit). Three zircon fractions from the Lake Shore Traps give an age of 1087.2 ± 1.6 Ma (68% probability of fit).

Data points for the baddeleyite analyses from the Portage Lake Volcanics are near concordia and broadly consistent with the zircon analyses. There is, however, an indication that the baddeleyite fractions may have slightly older $^{207}Pb/^{206}Pb$ ages. The oldest $^{207}Pb/^{206}Pb$ age is from fractured grains in the Copper City Flow which is also the most discordant data point (fraction 5, Table 1). Relatively discordant baddeleyite analyses which are slightly older than analyses from coexisting zircon fractions have already been observed in the Logan Sills [6] where the baddeleyite also formed thin wafer-like crystals. It was suggested that slight ^{222}Rn leakage from the ^{238}U decay chain, combined with secondary lead loss, may produce spuriously high $^{207}Pb/^{206}Pb$ ages. Rn loss may be facilitated due to the high surface area of the crystals. In other cases, such as prismatic baddeleyite from the Long Range Dikes of Labrador, U-Pb analyses of baddeleyite and zircon agree within experimental error [19]. Because of the possibility of systematic bias, the baddeleyite data have not been used in defining the ages. Since the baddeleyite must have formed in the mafic magma, the fact that it is no younger than the zircon demonstrates that there is no significant bias in the zircon ages due to inheritance.

The ages from the upper and lower parts of the Portage Lake Volcanics are resolved at the 1σ level and indicate a most probable time span of 2.2 ± 1.2 Ma (1σ error) for accumulation of the intervening lavas. The ages from the stratigraphically-higher Lake Shore Traps and the underlying Greenstone Flow are clearly resolved and indicate a time span of 6.8 ± 1.1 Ma (1σ error) for accumulation of the sediments and lavas in between.

The sandstone collected from the base of the Copper Harbor Conglomerate was from a heavy mineral-rich black sand lens containing abundant magnetite (now oxidized to hematite). The sample also contained abundant detrital zircon, mostly in the form of stubby crystals which show varying degrees of sedimentary abrasion from well-faceted to completely rounded. Analyses of three detrital zircon grains are on or near concordia and give $^{207}Pb/^{206}Pb$ ages between 1101 Ma and 1106 Ma (Fig. 4). A fraction consisting of 29 grains was also analyzed and gives a concordant data point with an age of 1104 ± 2 Ma.

3. Implications of age data

3.1. Volcanology

These geochronologic data, combined with stratigraphic and structural data, can be used to estimate limits on subsidence rates, lava repose periodicity and rates of magma production. These estimates are particularly important because the

general data base for precisely-dated flood basalt sequences is very limited. In addition, the data presented in this paper represent the first precisely-dated Precambrian flood basalts which can be compared to more recent examples of flood basalt volcanism which have received greater geochronological scrutiny.

Rates of graben subsidence can be calculated assuming that the surface upon which flood basalts were deposited remained balanced at a constant datum level. Flow morphology, very low primary slope variations [13] and distribution of interflow sediments provide supporting evidence for the maintenance of a nearly horizontal depositional surface (i.e. no volcanic edifice was constructed). Since subsidence may have exceeded lava flow deposition (resulting in a "starved" basin), calculated subsidence rates represent minimum values. An average thickness of 2850 m between the Greenstone and Copper City Flow horizons and the age difference of 2.2 ± 1.2 Ma (1σ error) yields a subsidence rate of $1.3^{+1.8}_{-0.6}$ mm/y. However, this value undoubtedly under-estimates peak subsidence rates within the central portion of the graben where seismic sections [20] show an axial fanning of dips and a greater accumulation of basalt (20 km) which presumably accumulated over the same time interval. If this is the case, the short duration of normal magnetic polarity lava accumulation (1098–1094 Ma) implies that peak subsidence rates in the central graben were about 5 mm/y. These rates are comparable to rates of tectonic subsidence in other continental grabens (typically 1–2 mm/y, range of 0.2–10 mm/y) [21–24].

The dated stratigraphic interval between the Greenstone Flow and the Lake Shore Traps (Fig. 2, average thickness of 1720 m, range of 1400–1850 m) yields considerably lower subsidence rates of 0.25 ± 0.07 mm/y. This value is considered a slight overestimate for rates of conglomerate deposition as the interval includes a substantial thickness of Portage Lake basalts. If the same subsidence rates calculated for the lower Portage Lake Volcanic stratigraphic interval apply to the lava flows above the Greenstone Flow, the adjusted subsidence rates during conglomerate deposition are lowered to 0.16 mm/y. This nearly tenfold difference is consistent with models of continental rift subsidence after active upwelling ceases. In the Keweenaw Peninsula, these two stages of subsidence were characterized by the preponderance of basalt volcanism and terrigenous clastic sedimentation, respectively.

The average repose period between eruptive events can also be estimated for the lower Portage Lake Volcanics stratigraphic interval. Up to 140 individual cooling units are recognizable between the Copper City and the Greenstone Flows. However, the presence of compound flows most likely reduces this number to 65–70 major eruptive events [25]. Therefore the average repose period between successive major eruptions was about 30 ka (16 ka, if 140 flows are assumed). Somewhat longer repose may typify eruptions separated by interflow sedimentary units while shorter repose periods may be typical of the majority of Portage Lake Volcanic flows which directly overlie subjacent flow tops.

Average rates of magma extrusion can also be calculated if the total volume of eruptive products produced over the lower dated interval is known. Unfortunately, present exposure precludes accurate volume estimates. A "minimum" volume of 40,000 km^3 for the Copper City Flow–Greenstone Flow interval is based on exposure of the Greenstone Flow on the Keweenaw Peninsula and Isle Royale and on the first-order assumption of constant thickness across the basin (Fig. 1C). A more "probable" volume of 100,000 km^3 (Fig. 1C) results from extending the area of the Portage Lake Volcanic basin based on geophysical data (i.e. fig. 4 of Green [26]). Assuming an age interval of 2.2 Ma, these volumes yield average rates of magma extrusion of 0.02 and 0.05 km^3/y respectively (about 10^{-4} km^3/y km along the rift). These rates fall within the range of values estimated for many other rift, hot spot and convergent volcanic settings (Iceland 0.05 km^3/y; Columbia River Basalts 0.07 km^3/y; Hawaii 0.03–0.1 km^3/y; central volcanos 0.009–0.03 km^3/y) [27,28].

However, recent high-precision geochronological investigations have indicated that continental flood basalt provinces may experience much higher rates of magma production compared with other magmatic settings (i.e. 0.28 km^3/y for Columbia River Basalts, and 0.4–1.0 km^3/y for Deccan Traps [29]). Similar high rates are obtained for the Midcontinent Rift if basalts outside the Portage Lake basin are included. A corresponding 3 km of basalt extruded over the same time interval

throughout the entire Lake Superior portion of the Midcontinent Rift yields a volume estimate of 420,000 km^3 and a production rate of about 0.2 km^3 /y (2.7×10^{-4} km^3/y km). The Greenstone Flow date of 1094.0 ± 1.5 Ma also implies that most normal magnetic polarity lavas, constituting > 90% of the exposed Midcontinent Rift section, were extruded over an approximate 4 Ma interval (1097.6 ± 3.7 Ma for the reverse magnetic polarity Osler Group [6]). If the volcanics buried in other portions of the Midcontinent Rift were also erupted over this short time span, average magma production rates may have been as high as 0.6 km^3/y. Although better volume data are clearly needed before accurate production rate estimates can be obtained, present Midcontinent Rift estimates are compatible with the high production rates predicted by models of mantle plume melting dynamics [30,31].

3.2. Sedimentation

The Copper Harbor Conglomerate, exposed on both the Keweenaw Peninsula and Isle Royale, is lithologically identical to the interflow sediments within the Portage Lake Volcanics, indicating that the terrigenous sediments had the same provenance [32]. Because of the high percentage of silicic volcanic debris in the conglomerates compared to the low number of felsic volcanic units within this portion of the Midcontinent Rift section, the source of these clasts has remained an enigma. Detrital zircon grains from a sandstone lens immediately overlying the Portage Lake Volcanics were analyzed in order to obtain some insight into the source rocks for the sediment. The ages from the three single detrital zircon grains (1101–1106 Ma, Fig. 4) correlate with lower Keweenawan rocks. The multigrain fraction was analyzed in order to search for any older component in the source. The fact that it is 1104 ± 2 Ma indicates that the provenance of the sediments is probably almost entirely lower Keweenawan in age.

Since the direction of sedimentary transport was from the margins toward the center of the Lake Superior basin [12,33], these coarse, terrigenous sediments could easily have been sampling uplifted early Keweenawan rocks exposed along rift flanks outside the graben border faults. Recent seismic reflection results [20] indicate that the Keweenawan Fault, and to a lesser degree the Isle Royale Fault, initially experienced movement in a normal sense. Therefore older, lower Keweenawan volcanic rocks, which were more likely to contain a greater proportion of felsic material relative to the Portage Lake Volcanics, were exposed to erosion along the rift flanks while younger basalt-dominated volcanic flows were accumulating within the central rift graben. Age patterns for detrital zircon from the Copper Harbor Conglomerate contrast with results from early Keweenawan conglomerate at the base of the Osler Group which indicate a significant component of earlier Proterozoic material in the source region [6].

3.3. Paleomagnetism

The age of the paleomagnetic field reversal is constrained between 1096.2 ± 1.8 Ma, the age of the normally polarized Copper City Flow, and 1097.6 ± 3.7 Ma, the age of the reversed part of the Osler Group [6]. The paleomagnetic polar wander curve (Fig. 5) shows a significant separation between reversed magnetic poles of different ages (1109–1098 Ma). Normal polarity poles, such as those for the Portage Lake Group, the Copper Harbor Conglomerate, the Lake Shore traps and the dated porphyry from Michipicoten Island [7] which intruded the Mamainse Point Formation (MP) and is overlain by the Michipicoten Island Formation (MI), are almost indistinguishable from each other. However, the age span which they represent (1096–1087 Ma) is almost as long as that represented by the reversed poles. The present age and paleomagnetic data are therefore inconsistent with a constant rate of paleomagnetic polar wander. This inconsistency could be resolved if there were an error in the pole for the Portage Lake Volcanics. There is a large structural correction applied to this pole, however an error in this correction would also be likely to bias the conformable Copper Harbor Conglomerate and Lake Shore Traps [34].

Halls and Pesonen [8] discussed possible reasons for the higher inclination of reversed relative to normal poles in the Keweenawan polar wander path. They concluded that a non-geocentric dipole during at least one polarity epoch was the best explanation for the major part of this reversal asymmetry. This effect may have been superim-

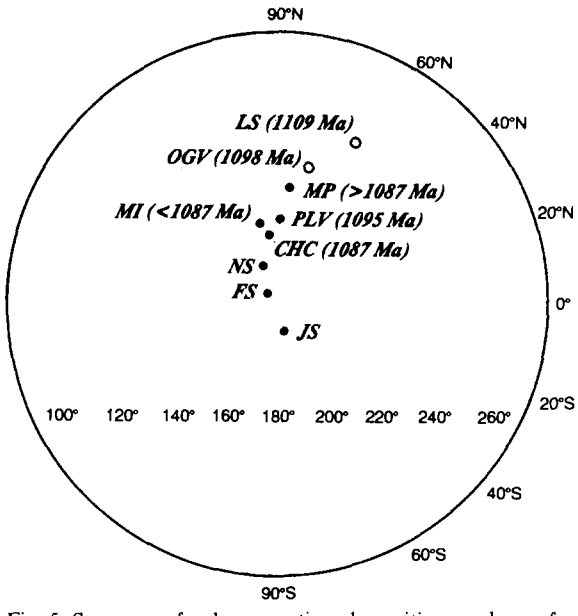


Fig. 5. Summary of paleomagnetic pole positions and ages for some Keweenawan units: *LS* = Logan Sills (average) [8,6], *OGV* = Osler Group Volcanics [35,6], *MP* = Mamainse Point Formation (exposed on Michipicoten Island) [7], *MI* = Michipicoten Island Formation [7], *PLV* = Portage Lake Volcanics [8], *CHC* = Copper Harbor Conglomerate [37], *NS* = Nonesuch Shale [37], *FS* = Freda Sandstone [37], *JS* = Jacobsville Sandstone [38]. Solid and open circles refer to normal and reversed poles, respectively.

posed on apparent polar wander due to relative motion of the continent with respect to the pole as shown by displacement of poles for progressively younger Keweenawan sediments (e.g. Freda Sandstone, Nonesuch Shale and Jacobsville Sandstone, Fig. 5). The noticeable displacement of poles between the upper part of the reversely magnetized Osler Group at 1097.6 ± 3.7 Ma and the lower part of the normally magnetized Portage Lake Volcanics at 1096.2 ± 1.8 Ma, despite the fact that the ages are within error of each other, supports the hypothesis of a non-geocentric dipole. The inconsistency in rates of apparent polar wander for normal relative to reversed poles can only be resolved by more extensive geochronology and precise paleomagnetism elsewhere in the Midcontinent Rift structure.

4. Conclusions

The ages of two basaltic units within the Portage Lake Volcanics (1096.2 ± 1.8 Ma and 1094.0 ± 1.5 Ma) and an andesite within the Copper Harbor Conglomerate (1087.2 ± 1.6 Ma) of the Keweenaw Peninsula are effectively resolved by U-Pb analyses of zircon. These samples represent a continuous stratigraphic thickness of 4.5 km. Measured age intervals indicate subsidence rates compatible with seismically observed rift-fill thickness and comparable to subsidence rates in other continental grabens. Precise U-Pb zircon dates now available indicate that while Midcontinent Rift magmatism spanned a 22 Ma time period from 1109 to 1087 Ma, the greatest portion of the exposed section was extruded over a much shorter, approximately 4 Ma time span between 1098 and 1094 Ma. Although age-volume estimates are not yet well defined, magma production rates calculated from the present data may have been similar to the high rates estimated for Mesozoic–Tertiary flood basalt provinces.

Lower Keweenawan ages (1101–1106 Ma) of detrital zircons in sandstones directly overlying the Portage Lake Volcanics indicate that rift-filling terrigenous sediments were largely derived from older Midcontinent Rift volcanic rocks exposed in uplifted flanks.

The age of the Keweenawan R–N paleomagnetic polarity reversal is well defined between 1097.6 ± 3.7 Ma and 1096.2 ± 1.8 Ma. There is a marked difference in polar wander rates defined using precisely dated reversed and normal poles. This suggests that the asymmetry between reversed and normal Keweenawan poles is largely the result of a non-geocentric dipole.

Acknowledgements

We would like to thank T.J. Bornhorst and W.I. Rose for numerous suggestions and helpful discussions during the early stages of this project. J.C. Green and H.C. Palmer read the manuscript and provided helpful suggestions. Sample crushing was carried out with the assistance of Jim Shaver. Mineral separation was by Sandra Kamo who also assisted with chemistry.

Part of this project was supported by a Natural Sciences and Engineering Research Council of Canada grant to the first author.

References

1 G.B. Morey and J.C. Green, Status of the Keweenawan as a stratigraphic unit in the Lake Superior region, in: Geol-

ogy and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 15–25, 1982.
2 W.R. Van Schmus and W.J. Hinze, The Midcontinent Rift system, Annu. Rev. Earth Planet. Sci. 13, 345–383, 1985.
3 W.R. Van Schmus, J.C. Green and H.C. Halls, Geochronology of Keweenawan rocks of the Lake Superior region: a summary, in: Geology and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 97–105, 1982.
4 L.T. Silver and J.C. Green, Time constants for Keweenawan igneous activity (abstract), Am. Geophys. Union Trans. (EOS) 44, 107, 1972.
5 T.E. Krogh, Improved accuracy of U-Pb ages by the creation of more concordant systems using an air abrasion technique, Geochim. Cosmochim. Acta 46, 637–649, 1982.
6 D.W. Davis and R.H. Sutcliffe, U-Pb ages from the Nipigon plate and northern Lake Superior, Geol. Soc. Am. Bull. 96, 1572–1579, 1985.
7 H.C. Palmer and D.W. Davis, Paleomagnetism and U-Pb geochronology of volcanic rocks from Michipicoten Island, Lake Superior, Canada: precise calibration of the Keweenawan polar wander track, Precambrian Res. 37, 157–171, 1987.
8 H.C. Halls and L.J. Pesonen, Paleomagnetism of Keweenawan rocks, in: Geology and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 173–201, 1982.
9 N.W.D. Massey, Keweenawan paleomagnetic reversals at Mamainse Point, Ontario: fault repetition or three reversals?, Can. J. Earth Sci. 16, 373–375, 1979.
10 K.W. Klewin and J.H. Berg, Evolution of the Midcontinent Rift: a perspective from the stratigraphy and chemistry of the Mamainse Point volcanics, Ontario (abstract), Am. Geophys. Union Trans. (EOS) 70, 273–274, 1989.
11 J.C. Green, Keweenawan plateau volcanism in the Lake Superior region, in: Volcanic Regimes in Canada, W.R.A. Barager, L.C. Coleman and J.M. Hall, eds., Geol. Assoc. Can., Spec. Pap. 16, 407–422, 1977.
12 P.A. Daniels Jr., Upper Precambrian sedimentary rocks: Oronto Group, Michigan–Wisconsin, in: Geology and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 107–133, 1982.
13 W.S. White, The Keweenawan lavas of Lake Superior, an example of flood basalts, Am. J. Sci. 258-A, 367–374, 1960.
14 J.C. Green, Geology of Keweenawan extrusive rocks, in: Geology and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 47–55, 1982.
15 H.R. Cornwall, Differentiation in lavas of the Keweenawan Series and the origin of the copper deposits of Michigan, Geol. Soc. Am. Bull. 62, 159–202, 1951.
16 T.E. Krogh, A low contamination method for hydrothermal decomposition of zircon and extraction of U and Pb for isotopic age determinations, Geochim. Cosmochim. Acta 37, 485–494, 1973.
17 D.W. Davis, C.E. Blackburn and T.E. Krogh, Zircon U-Pb ages from the Wabigoon–Manitou Lakes region, Wabigoon Subprovince, northwest Ontario, Can. J. Earth Sci. 19, 254–266, 1982.
18 D.W. Davis, Optimum linear regression and error estimation applied to U-Pb data, Can. J. Earth Sci. 19, 2141–2149, 1982.
19 S.L. Kamo, C.F. Gower and T.E. Krogh, A birthdate for the Iapetus Ocean? A precise U-Pb zircon and baddeleyite age for the Long Range dikes, southeast Labrador, Geology 17, 602–605, 1989.
20 W.F. Cannon, A.G. Green, D.R. Hutchinson, M. Lee, B. Milkereit, J.C. Behrendt, H.C. Halls, J.C. Green, A.B. Dickas, G.B. Morey, R. Sutcliffe and C. Spencer, The North American Midcontinent Rift beneath Lake Superior from GLIMPCE seismic reflection profiling, Tectonics 8, 305–332, 1989.
21 E. Groten, C. Gerstenecker and G. Hein, Geodetic high-precision measurements in active tectonic areas; example: the Rhinegraben, Tectonophysics 52, 157–165, 1979.
22 E. Tryggvason, Recent ground deformation in continental and oceanic rift zones, in: Continental and Oceanic Rifts, G. Palmason, ed., Am. Geophys. Union, Geodyn. Ser. 8, 17–29, 1982.
23 M.P. Golombek, G.E. McGill and L. Brown, Tectonic and geologic evolution of the Espanola Basin, Rio Grande Rift: structure, rate of extension, and relation to the state of stress in the western United States, Tectonophysics, 94, 483–507, 1983.
24 J.-M. Wang, The Fenwei rift and its recent periodic activity, Tectonophysics, 133, 257–275, 1987.
25 J.B. Paces, Magmatic processes, evolution and mantle source characteristics contributing to the petrogenesis of Midcontinent Rift basalts: Portage Lake Volcanics, Keweenawan Peninsula, Michigan, 413 pp., Ph.D. Thesis, Michigan Technological University, Houghton, Mich., 1988 (unpublished).
26 J.C. Green, Geological and geochemical evidence for the nature and development of the middle Proterozoic (Keweenawan) Midcontinent Rift of North America, Tectonophysics 94, 413–437, 1983.
27 D.A. Swanson, T.L. Wright and R.T. Helz, Linear vent systems and estimated rates of magma production and eruption for the Yakima basalt on the Columbia Plateau, Am. J. Sci. 275, 877–905, 1975.
28 G. Wadge, Steady state volcanism: evidence from eruption histories of polygenetic volcanoes, J. Geophys. Res. 87, 4035–4049, 1982.
29 A.K. Baksi, Elucidating the time of initiation and duration of volcanism for various Mesozoic–Tertiary flood basalt provinces (abstract), in: Continental Magmatism, Int. Assoc. Volcanol. Chem. Earth's Inter., General Assembly, Santa Fe, NM., p. 13, 1989.
30 R.S. White and D.P. Mackenzie, Magmatism at rift zones: the generation of volcanic continental margins and flood basalts, J. Geophys. Res. 94, 7685–7729, 1989.
31 M.A. Richards, R.A. Duncan and V.E. Courtillot, Flood basalts and hotspot tracts: plume heads and tails (abstract), in: Continental Magmatism, Int. Assoc. Volcanol. Chem. Earth's Inter., General Assembly, Santa Fe, NM., p. 222, 1989.
32 W.S. White, A theoretical basis for exploration for native copper in northern Wisconsin, U.S. Geol. Surv., Circ. 769, 19 pp., 1978.

33 G.P. Merk and M.A. Jirsa, Provenance and tectonic significance of the Keweenawan interflow sedimentary rocks, in: Geology and Tectonics of the Lake Superior Basin, R.J. Wold and W.J. Hinze, eds., Geol. Soc. Am., Mem. 156, 97–105, 1982.
34 H.C. Halls and H.C. Palmer, Remagnetization in Keweenawan rocks. Part II: lava flows within the Copper Harbour Conglomerate, Michigan, Can. J. Earth Sci. 18, 1395–1408, 1981.
35 H.C. Halls, A paleomagnetic reversal in the Osler volcanic group, northern Lake Superior, Can. J. Earth Sci. 11, 1200–1207, 1974.
36 T.D. Haig, A new paleomagnetic pole position for the interbedded lava flows within the Copper Harbour Conglomerate of Michigan's Keweenawan Peninsula, Ph.D. Thesis, Michigan Technological University, Houghton, Mich., 1986 (unpublished).
37 S.G. Henry, F.J. Mauk and R. Van der Voo, Paleomagnetism of the upper Keweenawan sediments: the Nonesuch Shale and Freda Sandstone, Can. J. Earth Sci. 14, 1128–1138, 1977.
38 J.L. Roy and W.A. Robertson, Paleomagnetism of the Jacobsville formation and the apparent polar wander path for the interval −1,100 to −670 m.y. for North America, J. Geophys. Res. 83, 1289–1304, 1978.
39 J.S. Stacey and J.D. Kramers, Approximation of terrestrial lead isotopic evolution by a two-stage model, Earth Planet. Sci. Lett. 26, 207–221, 1975.